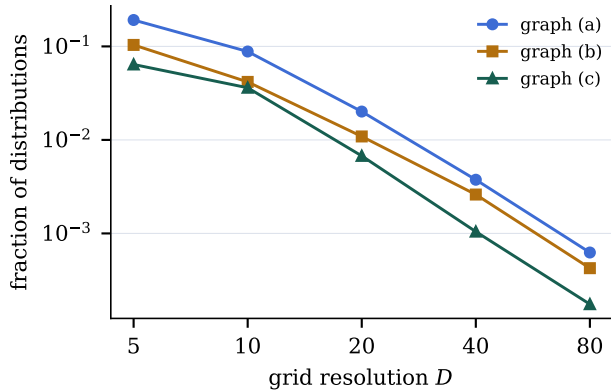


(i) partial-coarsening violations



(ii) union gap: $\Pi_{\cup} \subsetneq \Pi_{\text{do}(x)}$

